

Oluwatobi Iyanuoluwa Akinmuleya

oluwatobiiyanuoluwa1@gmail.com | github.com/tobai24 | tobai24.github.io

PROFESSIONAL SUMMARY

Medical doctor focused on data-efficient artificial intelligence for medical imaging, with emphasis on resource-constrained healthcare settings. Research interests include self-supervised learning, multimodal representation learning, and nuclear medicine imaging

EDUCATION

Shenyang Medical College

Bachelor of Medicine, Bachelor of Surgery (MBBS)

Shenyang, China

Oct. 2018 – Jul. 2025

RESEARCH EXPERIENCE

Research Supervisor (Connexin Team)

Remote

CAMERA MRI Africa – Medical Imaging Bootcamp

Sep. 2025 – Jan. 2026

- Supervised participants within the Connexin team on PET imaging analysis workflows, providing technical guidance on medical imaging pipelines, dataset interpretation, and reproducible analysis practices.

SPARK (MICCAI-Endorsed Academy for African Medical Imaging Translation)

Remote

Research Fellow

Jan. 2026 – Present

- Developed and evaluated deep learning approaches for medical imaging research, including leading a low-field MRI representation learning research and benchmarking brain MRI foundation models.
- Collaborated with interdisciplinary teams across radiology, medical physics, and computer science to advance clinically relevant AI methods for equitable medical imaging applications.

PUBLICATIONS & PRESENTATIONS

Publications

- Oluwatobi I. Akinmuleya**, Isaac Kofi Koomson, Cirubyankabagurhi Njuci Meschack-Augustin, Lucia Ndapewa Shambangu, MYRA: Multi-domain Hybrid MRI Representation Architecture for Self-Supervised Low-Field MRI, *MIRASOL 2026 (MICCAI 2026 Satellite Event)*, accepted.
- Amirmahdi Mojtahedzadeh, **Oluwatobi I. Akinmuleya**, Jeremiah Oluwatomi Itodo Daniel, Maxwell Adinavi, Region-Aligned Training and Target-Specific 2D–3D nnU-Net Fusion for BraTS 2026 Brain Metastasis Segmentation, *BraTS 2026*, accepted.
- Oluwatobi I. Akinmuleya**, Toufiq Musah, Improved Neuroanatomy Segmentation and Quality Assurance in Pediatric Ultra-Low-Field MRI, *MICCAI LISA Challenge 2026*, accepted.
- Oluwatobi I. Akinmuleya**, Olatokun Shamsudeen Akano, Samuel Danquah Ankapong, Olamide Lawal, and Toufiq Musah, Evaluating the Generalization of Neuroimaging Foundation Models to African Clinical MRI, *AFRICAI 2026 (MICCAI 2026 Satellite Event)*, accepted.
- B. Gheiji, S. Moradi, M. Vatanparast, S. Shokrani, M.A. Bagherzadeh, D. Elyassirad, **O.I. Akinmuleya**, S. Goodarzi, M. Heidari-Foroozan, M. Moassefi, J.D. Rudie, E. Calabrese, R. Jain, S. Faghani. NeuroAIHub: An AI-Driven Framework for Automated Curation and Discovery of Neuroradiology Datasets. *American Journal of Neuroradiology*, Accepted Manuscript, 2026.
- Oluwatobi I. Akinmuleya**, P.F. Cohen, K. Kairemo. 68Ga-PSMA PET/CT/MRI in the initial diagnosis and staging of prostate cancer: A review. *Advances in Radiotherapy & Nuclear Medicine*, Volume 2(4), 2024, 4590.

Presentations

- The Role of 68Ga-PSMA PET/CT and PET/MRI in the Diagnosis of Primary Prostate Cancer and TNM Staging**, Poster Presentation, International Conference of Hybrid Imaging, International Atomic Energy Agency (IAEA), Vienna (2024).

- **NeuroAI: An AI-Driven Framework for Automated Curation and Discovery of Neuroradiology Datasets**, Oral/Poster Presentation, American Society of Neuroradiology (ASNR) Annual Meeting, Philadelphia, USA (2025).
- **Training 105 African Clinicians and Students in Open Neuroimaging Analysis: CAMERA’s Dimensional Research Program**, Poster Presentation, International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting (2026).
- **Quantifying Amyloid Burden in Alzheimer’s Disease: A Reproducible Centiloid Pipeline for Analysis in Resource-Constrained Settings**, Poster Presentation, Alzheimer’s Association International Conference (AAIC) (2026).

SELECTED PROJECTS

In-Hospital Mortality Prediction System | *Python, MLflow, Docker, GCP, Streamlit, Evidently AI, Grafana*
Built an end-to-end machine learning pipeline for in-hospital mortality prediction to support clinical decision-making in critical care settings. Designed and deployed an MLOps system using MLflow, Docker, and GCP with integrated monitoring via Evidently AI and Grafana dashboards.

Computer Vision Systems for Medical and Surveillance Imaging | *PyTorch, YOLOv11, OpenCV, MTCNN*
Built real-time face recognition and tracking system using MTCNN and InceptionResNet for identity verification in video streams. Developed and fine-tuned YOLOv11 models for object detection in surveillance and gastrointestinal polyp detection for clinical imaging applications.

VOLUNTEER EXPERIENCE

Volunteer Tutor | *Shenyang Medical College, China* 2019 – 2023
Provided academic support and tutoring to medical students in core preclinical and clinical subjects, facilitating peer-to-peer learning and reinforcing foundational medical knowledge.

Tutorial Reviewer | *IndabaX Ghana (Ghana Data Science Summit)* 2025 – Present
Reviewed tutorial submissions for technical accuracy, clarity, and educational value, providing structured feedback to improve content quality and accessibility of data science education across diverse learner levels.

HONORS & AWARDS

- **Best Overall Medical Student**, Shenyang Medical College (2019–2021).
- **Academic Excellence Scholarship**, Shenyang Medical College (2019–2024).

SELECTED WORK EXPERIENCE

AI Engineer (Internship) May 2025 – Jul. 2025
Mnemos

- Developed AI-driven solutions to support medical education, contributing to tools for creating and enhancing learning resources for medical students across Africa.

Subinternship Feb. 2023 – May 2024
Greater Accra Regional Hospital *Accra, Ghana*

- Completed clinical subinternship training across multiple hospital departments, gaining hands-on experience in patient management and diagnostic workflows.
- Assisted in clinical case evaluation and supported decision-making processes under supervision of senior clinicians.
- Developed foundational clinical competence in hospital-based healthcare delivery within a resource-constrained setting.

SKILLS

Programming Languages: Python, JavaScript
Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, MONAI
Medical Imaging & Scientific Computing: nibabel, SimpleITK, OpenCV, torchvision
Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn
Web & API Development: Django, Flask, FastAPI, Streamlit, React, TailwindCSS, HTML/CSS, RESTful APIs
Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Kaggle